ME-2 Digital Twin Baseline Construction

Week 10: Temperature Drift Characterization and Compensation Inputs

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1 Purpose

The purpose of Week 10 is to experimentally characterize the effect of temperature on the structural sensors used by the behavioral digital twin.

Week 9 defined the digital twin outputs and identified temperature as an environmental conditioning variable in the expected-response relationship:

$$y_{\text{expected}} = f(L, E, T)$$

where:

- L represents applied loading,
- E represents vibration excitation,
- T represents environmental temperature.

Before temperature can be incorporated into the digital twin, the effect of temperature on the sensor measurements must first be quantified.

This stage therefore investigates whether the accelerometer and load-cell outputs change as the structure cools from an elevated temperature toward room temperature under otherwise nominally unchanged operating conditions.

The primary objectives are to:

- Measure sensor-output change during thermal cooldown,
- Identify temperature-dependent bias or drift,
- Separate long-term drift from transient measurement anomalies,

- Develop a MATLAB processing workflow for temperature-drift analysis,
- Determine whether temperature compensation should be included in the behavioral digital twin.

2 Relationship to the Digital Twin

The current digital twin compares measured structural behavior against an expected baseline.

The general residual relationship is:

$$r(t) = y_{\text{measured}}(t) - y_{\text{expected}}(t)$$

If sensor output changes significantly with temperature, a fixed baseline may incorrectly interpret normal environmental variation as an anomaly.

For example, the measured response may be represented more realistically as:

$$y_{\text{measured}} = y_{\text{structural}} + y_{\text{temperature}} + \epsilon$$

where:

- $y_{\text{structural}}$ is the response caused by the physical system,
- $y_{\text{temperature}}$ is the temperature-dependent sensor or system offset,
- ϵ represents random measurement noise.

The objective of the Week 10 study is therefore to begin separating temperature-dependent behavior from the response that should be interpreted as a structural anomaly.

A temperature-aware expected response may eventually take the form:

$$y_{\text{expected}} = f(L, E, T)$$

rather than:

$$y_{\text{expected}} = f(L, E)$$

This provides a more representative baseline for the AI-monitoring system.

3 Experimental Procedure

Temperature-drift testing was performed using the structural beam and existing sensing hardware.

The beam and sensor assembly were heated to approximately 60–65°C and then allowed to cool naturally toward room temperature.

Sensor measurements were recorded throughout the cooldown period.

Two datasets were collected:

1. Accelerometer measurements in the X, Y, and Z directions with corresponding temperature measurements.

2. Load-cell output with corresponding temperature measurements.

The objective was not to apply a changing mechanical input during this experiment. Instead, the cooldown itself was used as the changing environmental condition.

Any gradual change in sensor output observed during the cooldown can therefore be evaluated as a possible temperature-dependent response.

4 Experimental Data

The Week 10 temperature analysis uses two experimental datasets.

Table 1: Temperature Drift Experimental Data

File	Contents
`accelerometer_vib_tempvariation_821.xlsx`	Accelerometer X, Y, and Z measurements recorded simultaneously with beam temperature during thermal cooldown.
`load_cell_temperature_cleaned.xlsx`	Load-cell raw output recorded simultaneously with temperature during thermal cooldown.

The accelerometer dataset contains approximately 31,500 usable samples.

The recorded accelerometer temperature decreased from approximately:

$$60.25°\text{C}$$

to:

$$32.38°\text{C}$$

during the test.

For the load-cell dataset, the maximum recorded temperature was approximately:

$$65.56°\text{C}$$

and the cooldown continued to approximately:

$$32.88^{\circ}\mathrm{C}$$

These temperature ranges provide a sufficiently large environmental change to identify gradual sensor-output drift.

5 MATLAB Processing Workflow

A MATLAB processing script was developed to analyze the temperature-drift datasets.

The processing workflow consists of:

1. Reading the experimental Excel files,
2. Removing incomplete measurements,
3. Identifying and removing isolated measurement anomalies,
4. Identifying the beginning of the thermal cooldown,
5. Establishing an initial high-temperature sensor reference,
6. Smoothing high-frequency measurement noise,
7. Computing sensor drift relative to the initial reference,
8. Plotting sensor drift and temperature simultaneously.

The accelerometer and load-cell datasets are processed separately because the two sensor types exhibit different noise characteristics.

6 Accelerometer Drift Calculation

For each accelerometer axis, an initial reference value is established near the beginning of the cooldown.

The drift is calculated as:

$$\Delta a_x(t) = a_x(t) - a_{x,0}$$

$$\Delta a_y(t) = a_y(t) - a_{y,0}$$

$$\Delta a_z(t) = a_z(t) - a_{z,0}$$

where $a_{x,0}$, $a_{y,0}$, and $a_{z,0}$ represent the initial high-temperature baseline values.

The resulting values are converted from units of g to milli-g using:

$$1g = 1000 \text{ mg}$$

Therefore:

$$\Delta a_{\text{mg}} = 1000\,(a - a_0)$$

This representation makes relatively small changes in accelerometer bias easier to visualize.

7 Outlier and Anomaly Removal

Several large transient spikes were observed in the raw accelerometer dataset.

These spikes were substantially larger than the gradual response occurring during the thermal cooldown and were therefore treated as measurement anomalies rather than temperature drift.

The MATLAB workflow removes these transient events before smoothing and drift calculation.

A median-based outlier detector is used to identify measurements that differ significantly from the normal local signal behavior.

Detected outliers are replaced using interpolation between neighboring valid measurements.

This is important because allowing the transient spikes to remain in the signal would:

- Artificially increase the plotted drift range,
- Distort moving-average calculations,
- Affect baseline statistics,
- Potentially cause false anomaly classifications in the digital twin.

Isolated nonphysical temperature measurements were also removed prior to determining the beginning of the cooldown.

The cleaned signal therefore preserves the slow temperature-dependent behavior while rejecting short-duration acquisition artifacts.

8 Accelerometer Temperature-Drift Results

After anomaly removal and smoothing, the accelerometer data show a gradual change in all three measured axes as the beam cools.

The approximate high-temperature and low-temperature values are summarized below.

Table 2: Approximate Accelerometer Change During Cooldown

Axis	**High-Temperature Value**	**Low-Temperature Value**	**Approximate Change**
X	$0.11g$	$0.08g$	-30 mg
Y	$0.14g$	$0.13g$	-10 mg
Z	$0.86g$	0.90–$0.91g$	$+40$–50 mg

The direction of the drift is not identical for each axis.

The X and Y measurements decrease as temperature decreases, while the Z measurement increases.

The largest observed change occurs in the Z axis.

These results indicate that temperature has a measurable effect on the accelerometer output over the tested temperature range.

However, the current experiment measures the response of the complete sensor/beam/test configuration. Therefore, the observed change may contain contributions from both sensor bias and small temperature-induced physical changes in the structure or mounting configuration.

The primary conclusion at this stage is that the measured acceleration baseline is not completely temperature independent.

9 Accelerometer Drift Plot

Figure 1 shows the temperature history together with the calculated accelerometer drift.

10 Accelerometer Drift Plot

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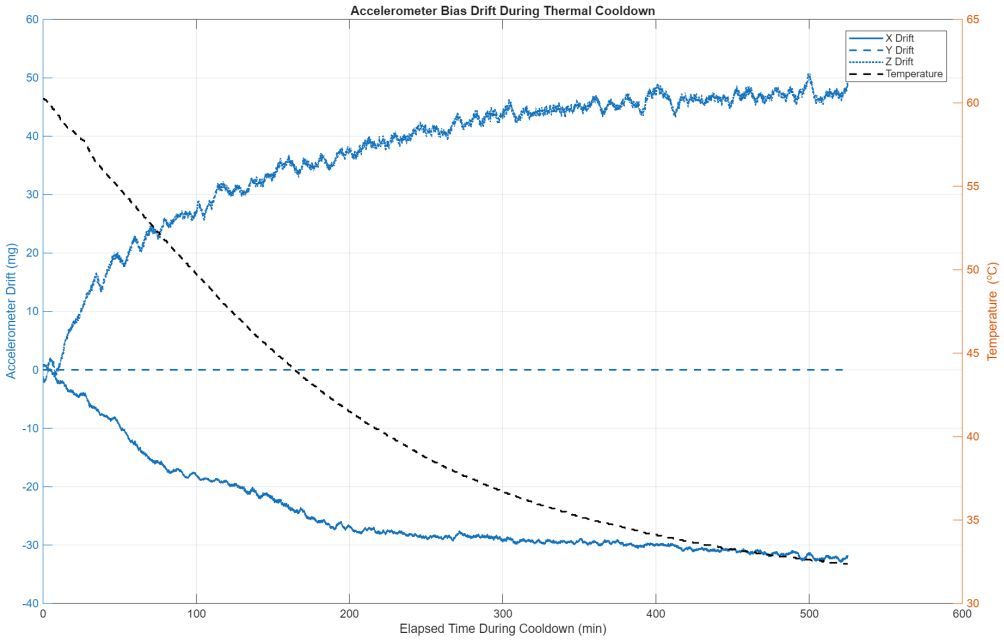


Figure 1: Accelerometer bias drift during thermal cooldown.

The expected thermal behavior is a gradual change in the baseline rather than a short-duration excursion.

This distinction is important for digital-twin anomaly detection because the system should respond differently to gradual environmental drift and sudden abnormal sensor events.

11 Load-Cell Temperature Drift

The load-cell dataset was analyzed using a similar process.

The initial load-cell reference is defined as:

$$L_0$$

and the load-cell drift is:

$$\Delta L(t) = L(t) - L_0$$

Because the current dataset contains raw load-cell output rather than calibrated force units, the drift is currently represented in raw counts.

A moving median was used to suppress short-duration spikes while preserving the slower temperature-dependent response.

The load-cell output showed a clear change during the cooldown.

Using representative median values near the beginning and end of the test, the output increased from approximately:

$$466{,}000$$

raw counts near the high-temperature condition to approximately:

$$482{,}000$$

raw counts near the low-temperature condition.

This corresponds to an approximate change of:

$$\Delta L \approx 15{,}000$$

raw counts over the tested cooldown range.

The direction of the relationship indicates that the load-cell output increased as temperature decreased.

12 Load-Cell Drift Plot

Figure 2 shows the cleaned load-cell response and temperature history.

13 Accelerometer Drift Plot

Figure 1 shows the temperature history together with the calculated accelerometer drift.

14 Load-Cell Drift Plot

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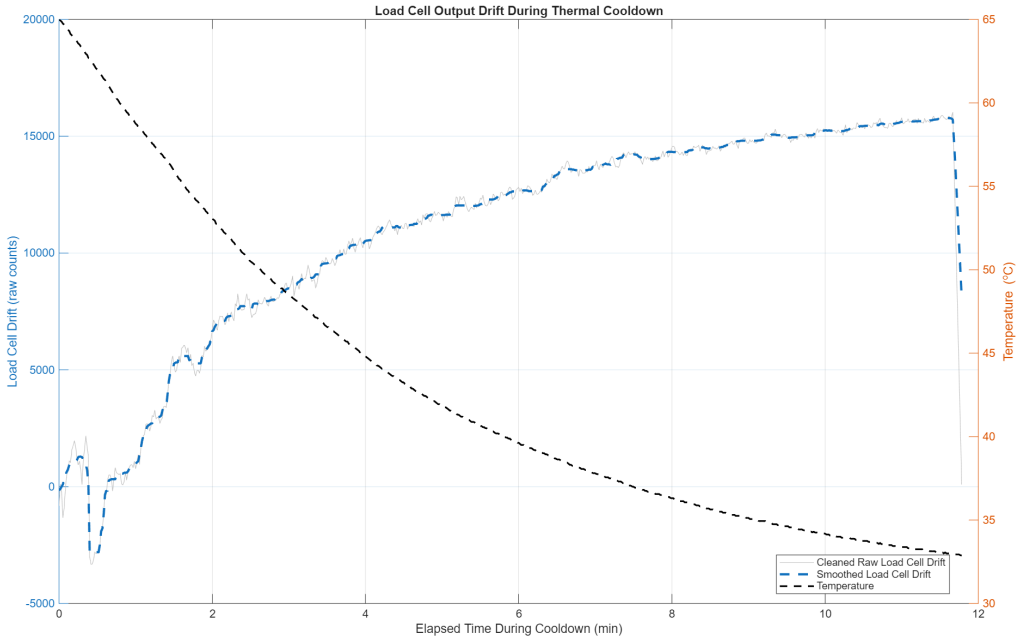


Figure 2: Load-cell output drift during thermal cooldown.

The smoothed load-cell response follows a substantially more gradual trend than the isolated raw measurement spikes.

This supports the use of temperature as an explicit input to future load-cell baseline models.

15 Implications for the Digital Twin

The temperature-drift experiment demonstrates that the structural sensor baseline is not constant across the tested temperature range.

This has direct implications for the residual calculation used by the digital twin.

If the expected response is assumed to be constant:

$$y_{\text{expected}} = y_0$$

then the residual becomes:

$$r(t) = y_{\text{measured}}(t) - y_0$$

A temperature-dependent sensor offset could then produce a large residual even when no actual structural fault is present.

A more appropriate model is:

$$y_{\text{expected}} = f(L, E, T)$$

so that the expected response changes with environmental temperature.

A first-order temperature correction could eventually be represented as:

$$y_{\text{corrected}} = y_{\text{measured}} - k_T(T - T_{\text{ref}})$$

where:

- T_{ref} is the reference temperature,
- k_T is the experimentally determined temperature-drift coefficient.

The Week 10 dataset provides the experimental information needed to begin estimating these coefficients.

16 AI and Anomaly-Detection Relevance

Temperature compensation is especially important for the AI-monitoring portion of the digital twin.

Without environmental context, the AI system may interpret normal sensor drift as:

- Structural degradation,
- Abnormal loading,
- Sensor malfunction,
- Sensor spoofing,
- Unexpected behavioral deviation.

A temperature-aware baseline allows the system to distinguish between:

Environmental Drift

and:

True Behavioral Anomaly

This reduces the likelihood of false-positive anomaly detections.

Temperature can therefore be transmitted alongside the sensor and twin-output data so that downstream AI models receive both the measured response and the environmental state associated with that response.

17 Recommended Temperature-Aware Twin Outputs

Based on the Week 10 results, temperature should be added directly to the structured digital-twin output.

Table 3: Recommended Temperature-Aware Twin Outputs

Field	**Type**	**Purpose**
Timestamp	Float	Synchronizes measurements with telemetry
Temperature	Float	Provides current environmental condition
Measured Response	Float	Raw or calibrated sensor measurement
Temperature-Corrected Response	Float	Response after environmental compensation
Expected Response	Float	Temperature-conditioned twin prediction
Residual	Float	Difference between corrected measured and expected behavior
Residual RMS	Float	Long-window deviation metric
Temperature Drift Estimate	Float	Estimated sensor offset associated with temperature
Bound Violation Flag	Boolean	Indicates operation outside expected behavioral limits

18 Next Development Stage

The current Week 10 results establish that temperature-dependent behavior is measurable.

The next stage should focus on converting the observed trends into usable compensation models.

Recommended next steps include:

- Repeat the thermal test to evaluate repeatability,
- Perform multiple heating and cooling cycles,
- Evaluate whether heating and cooling produce hysteresis,
- Determine temperature-drift coefficients for each accelerometer axis,

- Convert load-cell raw counts to calibrated force units,
- Determine the load-cell temperature coefficient in engineering units,
- Incorporate temperature into the expected-response model,
- Compare residuals before and after temperature compensation.

Repeated trials will also help determine whether the observed trends are caused primarily by the sensors themselves or by temperature-dependent behavior of the complete structural assembly.

19 Summary

Week 10 experimentally evaluates temperature drift in the accelerometer and load-cell measurements used by the behavioral digital twin.

The beam and sensor assembly were heated to approximately 60–65°C and allowed to cool toward approximately 33°C while sensor measurements were recorded.

MATLAB processing was developed to:

- Import the experimental datasets,
- Remove measurement anomalies,
- Isolate the cooldown period,
- Smooth high-frequency noise,
- Establish high-temperature reference values,
- Calculate sensor-output drift,
- Compare sensor response directly against temperature.

The accelerometer measurements exhibited approximately:

$$-30 \text{ mg}$$

of change in the X axis,

$$-10 \text{ mg}$$

in the Y axis, and approximately:

$$+40\text{–}50 \text{ mg}$$

in the Z axis across the tested temperature range.

The load-cell output also exhibited a clear temperature-dependent trend, changing by approximately:

15,000

raw counts between the high- and low-temperature operating conditions.

These results indicate that temperature should be treated as a meaningful conditioning variable within the behavioral digital twin.

The Week 10 work therefore provides the experimental basis for future temperature compensation and should reduce false anomaly classifications caused by normal environmental variation.

A Appendix A: Associated Project Files

The following files support the Week 10 temperature-drift study.

Table 4: Associated Temperature Drift Files

File	Purpose
`accelerometer_vib_tempvariation_821.xlsx`	Experimental accelerometer and temperature dataset collected during thermal cooldown.
`load_cell_temperature_cleaned.xlsx`	Experimental load-cell and temperature dataset collected during thermal cooldown.
`temperature_drift_analysis.m`	MATLAB implementation used for anomaly removal, smoothing, reference calculation, drift analysis, and visualization.
`accelerometer_temperature_drift.png`	Processed accelerometer temperature-drift visualization.
`load_cell_temperature_drift.png`	Processed load-cell temperature-drift visualization.

All experimental data, MATLAB processing implementations, and associated figures should be maintained with the digital-twin project files for reproducibility and future temperature-compensation development.